

what we can do with it. Whether you are a developer, a designer, a decision-maker, or just someone curious about the future, now is the time to lean into GenAI. Because the next big thing is not just smart, it is creative.

Think about Siri. Yes, the friendly voice assistant on your phone that patiently answers your questions (and sometimes dodges them). When we speak to Siri, what we are experiencing is the magic of artificial intelligence—machines trying to mimic human behaviour. AI's ultimate goal? To think, respond, and learn like us. From Siri mimicking human interaction to self-learning thermostats, from Netflix recommending your next binge to AI writing your next favourite novel—machine intelligence is becoming less artificial and more authentic.

AI is the umbrella term, and under it sits a key player: Machine Learning (ML). In ML, we do not need to write condition-based code like 'if this, then that.' Instead, machines learn from patterns and data. Think of it like teaching a child by example rather than strict rules: "1 + 1 = 2" or "The sky is blue." Feed the system enough data, and it begins to predict outcomes. That is ML in action.

Take IoT (Internet of Things), for example. Our homes are already dotted with smart devices quietly learning from our habits. A geyser knows when we typically wake up. An air-conditioner remembers what temperature we like at bedtime. Based

### Tail-end benefits: Why GenAI is not just tech hype

Let's not forget the end benefits that make GenAI so compelling:

- 30-32% overall productivity increase
- Driver and road safety improvements
- Smarter traffic management
- Fuel economy optimisation
- Personalised in-cabin experiences
- Faster emergency response systems

So, yes—GenAI isn't just a buzzword. It is the nervous system of next-gen automotive and manufacturing operations.

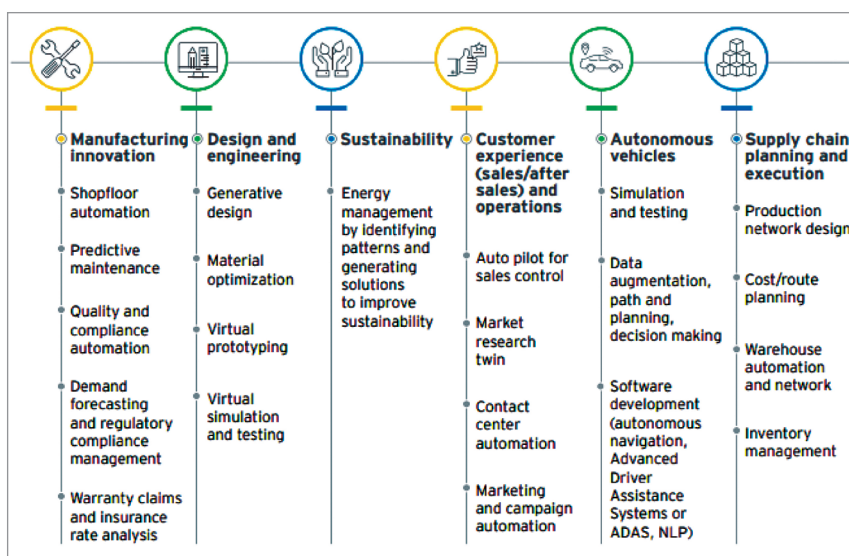


Fig. 2: Automotive and mobility value chain

on this learning, the system might soon ask, "Should I turn on the AC to 24°C now?" That is predictive ML working behind the scenes—anticipating needs, not just reacting to them.

### Predict vs create

While machine learning is great at predicting—say, recommending the next film on Netflix based on your viewing history—Generative AI goes one step further: it creates. Yes, it creates from scratch. It could be an image, a song, a video, or even an entire essay. It learns from unstructured data—messy, complex inputs that do not sit neatly in spreadsheets. Think pixel-rich images, streaming videos, flowing human text—nothing in neat rows and columns.

Now, how does it do that? The secret lies in deep learning, a subfield of ML. Deep learning uses neural networks modelled after the human brain. Imagine layers of "neurons" processing data deeply—hence the name. Within deep learning, the Transformer architecture emerged. This is the engine powering modern generative models—from ChatGPT to DALL·E to those AI-generated Drake songs. There are two primary types of machine learning:

**1. Statistical ML.** Works on structured data (like Excel sheets).

Algorithms include linear regression, random forests, and gradient boosting

**2. Deep learning.** Works on unstructured data (like images or audio). It uses neural networks and excels with large datasets

Generative AI lives in the deep learning world, thriving on massive datasets and building its own creative output, whether that is a poem or a hyper-realistic face that does not exist.

### Trends in the automobile sector

Now let us shift gears—quite literally—to the automobile industry. The biggest trend? Electrification and autonomy. EVs (electric vehicles) are no longer futuristic dreams; they are rolling off assembly lines at an accelerating pace. With improved battery technologies, better charging infrastructure, and a global push for sustainability, EV adoption is booming. By 2026, many analysts predict a significant milestone in EV penetration worldwide.

But where does AI fit in? Right at the heart of it. AI and machine learning power:

- Battery management systems that predict performance and lifespan
- Charge station management systems (CSMS) optimising the grid
- Driver behaviour analytics to tailor in-car experiences